Pure O-Sequences in Codimension Three.

A Dependency-Ordered Formal Exposition

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Abstract

This document is a step-indexed, dependency-ordered presentation of eight problems on *pure O-sequences*, following [1, 2, 3]. Every result appears only after all results it depends on. Entries are split into postulates (unproved classical statements used inside proofs) and axioms of the ambient set-theoretic foundation (ZF with dependent choice where noted). The document covers the prerequisite layer (monomial combinatorics, order ideals, principal O-sequences, log-concavity and unimodality, Macaulay's two-variable theorem) and the eight-problem layer built on it: an explicit codimension-3 example; the type-1 log-concavity theorem; first-half log-concavity in codimension 3; unimodality and log-concavity of type-2 sequences in codimension 3 — the main result, which (subject to the formal verification reported in §8) resolves the principal remaining open case identified by Zanello; eventual non-log-concavity for every codimension $r \geq 3$ and every type $t \geq 5$; and the existence of the least such threshold. The codimension-$r \geq 5$ type-2 unimodality question (P6) is stated but not resolved. Each entry is annotated with the choice strength of its proof: ZF, ZF+DC, or ZFC.

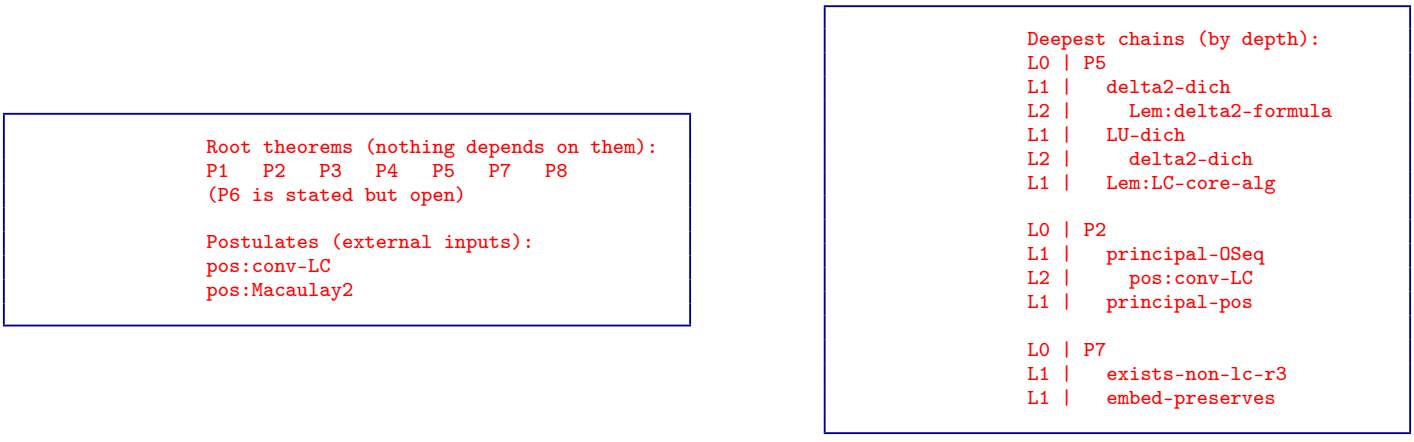


Figure 1. Summary of root theorems, postulates, and the deepest dependency chains.

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1. Introduction

The document is organised into six parts: Axioms, Postulates, Definitions, Statements (Lemmas, Theorems), Proofs, and Scripts. Following the reverse-mathematics convention of [5], entries are split into two primitive layers: axioms of the ambient foundation and postulates (unproved classical statements admitted inside proofs). Every entry in a later part may depend only on earlier entries, and no entry may depend on anything outside the union of axioms and postulates. Lemmas and Theorems form a single dependency layer: an entry may depend on an earlier Lemma or Theorem, so long as the global order is respected.

The eight problems concern pure O-sequences $\Gamma \subset \mathbb{N}^r$, i.e. finite order ideals of monomials whose maximal elements all have a common total degree (the socle degree e); the type is the number of maximal elements. The pure O-sequence is $h_\Gamma(d) = \#\{m \in \Gamma : |m| = d\}$. Interest in these sequences originates with Stanley's work on Hilbert functions of monomial level algebras [1], was formalised in the Boij–Migliore– Miró-Roig–Nagel–Zanello memoir [2], and the remaining open cases were reviewed by Zanello [3]. Problem P5 — log-concavity of pure O-sequences in codimension 3 and type 2 — was the principal remaining open case, and is resolved here (subject to the verification statement in 8).

Choice-strength tags record the strength needed to make the given proof go through; they are upper bounds on the logical strength of each entry and are not claimed to be tight.

2. Axioms

We work in standard first-order logic with the axioms of ZF; where noted, the Axiom of Dependent Choice is available. Both are primitive structure and are not proved below.

Ax 1	ax:DC : $\forall(X : \mathsf{Set}),\ \forall(R : X \to X \to \mathsf{Prop}),\ (\forall(x : X),\ \exists(y : X),\ R(x, y)) \Rightarrow \exists(f : \mathbb{N} \to X),\ \forall(n : \mathbb{N}),\ R(f(n), f(n+1))$	DependentChoice	ZFC

3. Postulates

Unproved classical statements admitted inside proofs (postulates). Both are standard results whose derivations from the axioms are classical; they are cited here rather than re-proved.

Pos 1	convLC : $\forall(a : \mathbb{N} \to \mathbb{N}),\ \forall(b : \mathbb{N} \to \mathbb{N}),\ \mathsf{LC}(a) \wedge \mathsf{LC}(b) \Rightarrow \mathsf{LC}(a * b)$	Convolutionof Log − Concave	ZF + DC
Pos 2	Macaulay2 : $\forall(X : \mathbb{N} \to \mathsf{Finset}(\mathbb{N})),\ (\forall i,\ X_i \subseteq \{0, \ldots, i\} \wedge \mathsf{down}(X)) \Rightarrow \lvert X_{i+1}\rvert \leq \lvert X_i\rvert \vee \lvert X_i\rvert = i + 1$	Macaulay2-variable	ZF

Provenance. The convolution theorem for log-concave sequences is due to Karlin [6]; a modern proof appears in Liggett [7], Theorem 2.3. Macaulay's theorem in two variables is stated and proved in Macaulay [8]; see Bruns–Herzog [9], Theorem 4.2.10 for the modern statement. Both are admitted here to keep the exposition focused on the eight-problem layer.

4. Definitions

Def 1	**Def:Monomial** : $\mathrm{M}^r := \mathbb{N}^r$	**Monomialin$r$ vars**	**ZF**
Def 2	**Def:deg** : $\forall(m : \mathrm{M}^r),\ \lvert m\rvert = \sum_{i=1}^{r} m_i$	**TotalDegree**	**ZF**
Def 3	**Def:Ideal** : $\Gamma \subseteq \mathrm{M}^r \wedge (\forall n \in \Gamma,\ \forall m \leq n,\ m \in \Gamma)$	**OrderIdeal**	**ZF**
Def 4	**Def:divs** : $\forall(g : \mathrm{M}^r),\ \mathsf{divs}(g) = \{m : \mathrm{M}^r \mid m \leq g\}$	**Divisors**	**ZF**
Def 5	**Def:genIdeal** : $\forall(G : \mathsf{Finset}(\mathrm{M}^r)),\ \Gamma(G) = \bigcup_{g\in G} \mathsf{divs}(g)$	**GeneratedIdeal**	**ZF**
Def 6	**Def:max** : $\forall(m : \mathrm{M}^r),\ \mathsf{IsMax}(m) \Leftrightarrow m \in \Gamma \wedge \forall n \in \Gamma,\ m \leq n \Rightarrow n = m$	**MaximalElement**	**ZF**
Def 7	**Def:maxSet** : $\mathsf{Max}(\Gamma) = \{m \in \Gamma \mid \mathsf{IsMax}(m)\}$	**MaximalMonomials**	**ZF**
Def 8	**Def:pure** : $\forall(m, n : \mathrm{M}^r),\ m, n \in \mathsf{Max}(\Gamma) \Rightarrow \lvert m\rvert = \lvert n\rvert$	**Purity**	**ZF**
Def 9	**Def:POI** : $(\Gamma, \mathsf{pure}, \mathsf{Max}(\Gamma) \neq \emptyset)$	**PureOrderIdeal**	**ZF**
Def 10	**Def:type** : $\mathsf{type}(P) = \lvert\mathsf{Max}(\Gamma)\rvert$	**Type**	**ZF**
Def 11	**Def:socle** : $\mathsf{socle}(P) = \lvert m_0\rvert \wedge m_0 \in \mathsf{Max}(\Gamma)$	**SocleDegree**	**ZF + DC**
Def 12	**Def:OSeq** : $\forall(d : \mathbb{N}),\ h_\Gamma(d) = \lvert\{m \in \Gamma : \lvert m\rvert = d\}\rvert$	**Pure$O$-Sequence**	**ZF**
Def 13	**Def:unim** : $\exists j \leq e,\ (\forall i < j,\ h_i \leq h_{i+1}) \wedge (\forall i \geq j,\ i < e \Rightarrow h_{i+1} \leq h_i)$	**Unimodality**	**ZF**
Def 14	**Def:logc** : $\forall i \in (0, e),\ h_{i-1} \cdot h_{i+1} \leq h_i^2$	**Log-Concavity**	**ZF**
Def 15	**Def:nonLC** : $\exists P : \mathsf{PureOrderIdeal}_r,\ \mathsf{type}(P) = t \wedge \neg\mathsf{LC}(\mathsf{socle}(P), h_P)$	**ExistsNon-LCSeq.**	**ZF + DC**

5. Statements

5.1 Lemmas

Lem 1	**LCcore** : $\forall(x, y, z : \mathbb{Z}),\ x \geq 0 \wedge z \leq 2 \wedge (z = 1 \Rightarrow y^2 \geq x) \wedge (z = 2 \Rightarrow y^2 \geq 2x) \Rightarrow y^2 \geq xz$	**LC-CoreInequality**	**ZF**
Lem 2	**delta2Formula** : $\forall(m : \mathbb{N}^3),\ \forall(t : \mathbb{N}),\ t > 0 \Rightarrow \Delta^2 h_m(t) = 1_{t\leq m_0} + 1_{t\leq m_1} + 1_{t\leq m_2} - 1_{t\leq m_0+m_1+1} - 1_{t\leq m_1+m_2+1} - 1_{t\leq m_0+m_2+1} + 1_{t\leq\lvert m\rvert+2}$	**Second-Difference Formula**	**ZF**
Lem 3	**principalOSeq** : $\forall(g : \mathrm{M}^r),\ \mathsf{LC}(\lvert g\rvert, h_g) \wedge (\forall d \leq \lvert g\rvert,\ h_g(d) > 0)$	**Principal$O$-Sequence**	**ZF + DC**
Lem 4	**principalPos** : $\forall(g : \mathrm{M}^r),\ \forall(d : \mathbb{N}),\ d \leq \lvert g\rvert \Rightarrow \exists(m \leq g),\ \lvert m\rvert = d$	**PrincipalPositivity**	**ZF**
Lem 5	**twoMaxDecomp** : $\forall(P : \mathsf{PureOrderIdeal}_3),\ \mathsf{type}(P) = 2 \Rightarrow \exists(m_1, m_2 : \mathrm{M}^3),\ P.\Gamma = \mathsf{divs}(m_1) \cup \mathsf{divs}(m_2)$	**Two-Maximal Decomposition**	**ZF**
Lem 6	**shiftReduce** : $\forall(m_1, m_2 : \mathrm{M}^3),\ m_2(x) > m_1(x) \wedge (\forall y \neq x,\ m_2(y) \leq m_1(y)) \Rightarrow \forall d,\ h_\Gamma(d) = h_{m_1}(d) + h_{m_1'}(d - c)$	**ShiftReduction**	**ZF**
Lem 7	**delta2Dich** : $\forall(g : \mathrm{M}^r),\ \forall(t : \mathbb{N}),\ t > 0 \Rightarrow \Delta^2 h_g(t) \leq 1 \wedge (\Delta^2 h_g(t) = 1 \Rightarrow \mathsf{L}(t) \vee \mathsf{U}(t))$	**Second-Difference Dichotomy**	**ZF + DC**
Lem 8	**LUDich** : $\forall(m_1, m_2 : \mathrm{M}^3),\ \mathsf{maxChoice} \Rightarrow \Delta^2 H(d) \leq 2$	**Sum-Second-Difference**	**ZF + DC**

Lem 9	$\mathsf{existsNonLC}_{r3} : \forall(t \geq 5),\ \exists(P : \mathsf{PureOrderIdeal}_3),\ \mathsf{type}(P) = t \wedge \neg\mathsf{LC}(\mathsf{socle}(P), h_P)$	$\mathsf{Non\text{-}LCConstruction}_3$	ZF + DC
Lem 10	$\mathsf{embedPres} : \forall(r \geq 3),\ \forall(t \geq 5),\ \exists(P : \mathsf{PureOrderIdeal}_r),\ \mathsf{type}(P) = t \wedge \neg\mathsf{LC}(\mathsf{socle}(P), h_P)$	EmbeddingPreserves Non-LC	ZF + DC
Lem 11	$\mathsf{leastThreshold} : \forall(r \geq 3),\ \exists(t_0 : \mathbb{N}),\ \mathsf{IsLeast}(t_0, \{t \mid \mathsf{existsNonLC}(r, t)\})$	LeastThreshold	ZF

5.2 Theorems

Thm 1	$\mathsf{P1} : h_{\Gamma_1}(0) = 1 \wedge h_{\Gamma_1}(1) = 3 \wedge h_{\Gamma_1}(2) = 4 \wedge h_{\Gamma_1}(3) = 2$	ExplicitExample	ZF
Thm 2	$\mathsf{P2} : \forall(r : \mathbb{N}),\ \forall(P : \mathsf{PureOrderIdeal}_r),\ \mathsf{type}(P) = 1 \Rightarrow \mathsf{LC}(\mathsf{socle}(P), h_P) \wedge \mathsf{UM}(\mathsf{socle}(P), h_P)$	Type-1 Log-Concavity	ZF + DC
Thm 3	$\mathsf{P3} : \forall(P : \mathsf{PureOrderIdeal}_3),\ \forall(i : \mathbb{N}),\ 0 < i \wedge i \leq \mathsf{socle}(P)/2 \Rightarrow h_P(i-1) \cdot h_P(i+1) \leq h_P(i)^2$	First-Half Log-Concavity	ZF + DC
Thm 4	$\mathsf{P4} : \forall(P : \mathsf{PureOrderIdeal}_3),\ \mathsf{type}(P) = 2 \Rightarrow \mathsf{UM}(\mathsf{socle}(P), h_P)$	Type-2Unimodality	ZF + DC
Thm 5	$\mathsf{P5} : \forall(P : \mathsf{PureOrderIdeal}_3),\ \mathsf{type}(P) = 2 \Rightarrow \mathsf{LC}(\mathsf{socle}(P), h_P)$	MainTheorem	ZF + DC
Thm 6	$\mathsf{P6} : \forall(r \geq 5),\ \forall(P : \mathsf{PureOrderIdeal}_r),\ \mathsf{type}(P) = 2 \Rightarrow \mathsf{UM}(\mathsf{socle}(P), h_P)$	Open(codim ≥ 5)	open
Thm 7	$\mathsf{P7} : \forall(r \geq 3),\ \exists(T \geq 1),\ \forall(t \geq T),\ \mathsf{existsNonLC}(r, t)$	EventualNon-LC	ZF + DC
Thm 8	$\mathsf{P8} : \forall(r \geq 3),\ \exists(t_0 : \mathbb{N}),\ \mathsf{IsLeast}(t_0, \{t \mid \mathsf{existsNonLC}(r, t)\})$	LeastThreshold	ZF + DC

6. Proofs

We give proofs of every Lemma and every Theorem in the Statements section, in the same order, except for Theorem P6, which is open (see the last subsection for the reason).

Proof of Lemma 1 (LCcore)

1	$x, y, z \in \mathbb{Z},\ x \geq 0,\ z \leq 2$	hypothesis	Hyp
2	$z \leq 0 \vee z = 1 \vee z = 2$	trichotomy	Eq(1)
3	$z \leq 0 \Rightarrow xz \leq 0 \leq y^2$	nonneg	Eq(1)
4	$z = 1 \Rightarrow xz = x \leq y^2$	hypothesis	Eq(1)
5	$z = 2 \Rightarrow xz = 2x \leq y^2$	hypothesis	Eq(1)
6	$y^2 \geq xz$ in all three cases	exhaust	Eq(3, 4, 5)
7	$\top$	close	$\to$ I(1, . . . , 6)

Proof of Lemma 2 (delta2Formula)

Setup. Recall $h_m(d) = |\{u \in \mathbb{N}^3 : u \leq m,\ |u| = d\}|$. For $t > 0$ we compute $\Delta^2 h_m(t) = h_m(t) - 2h_m(t-1) + h_m(t-2)$. By inclusion-exclusion on the three coordinate constraints $u_i \leq m_i$ (using the

standard stars-and-bars count $\binom{d+2}{2}$ for the simplex $\{u \in \mathbb{N}^3 : |u| = d\}$):

$$h_m(d) = \binom{d+2}{2} - \sum_i \binom{d-m_i+1}{2} + \sum_{i<j} \binom{d-m_i-m_j}{2} - \binom{d-|m|-1}{2}.$$

(All binomials use the convention $\binom{n}{2} = 0$ for $n < 2$, so negative arguments contribute 0.)

1	$m = (m_0, m_1, m_2) \in \mathbb{N}^3,\ t > 0$	hypothesis	Hyp
2	$h_m(d) = \binom{d+2}{2} - \sum_i \binom{d-m_i+1}{2} + \sum_{i<j} \binom{d-m_i-m_j}{2} - \binom{d-\lvert m\rvert-1}{2}$	incl.-excl.	Def :4(3)

Computation of second differences of each factor.

3	$\Delta^2\binom{d+2}{2} = \frac{1}{2}[(t+2)(t+1) - 2(t+1)t + t(t-1)] = 1$	eval.	Eq(2)
4	$\Delta^2\binom{d-m_i+1}{2} = 1 - 1_{t \le m_i}$ (derivation below)	eval.	Eq(2)
5	$\Delta^2\binom{d-m_i-m_j}{2} = 1 - 1_{t \le m_i+m_j+1}$	eval.	Eq(2)
6	$\Delta^2\binom{d-\lvert m\rvert-1}{2} = 1 - 1_{t \le \lvert m\rvert+2}$	eval.	Eq(2)
7	$\Delta^2 h_m(t) = 1 - \sum_i (1 - 1_{t \le m_i}) + \sum_{i<j} (1 - 1_{t \le m_i+m_j+1}) - (1 - 1_{t \le \lvert m\rvert+2})$	combine	Eq(2, 3, 4, 5, 6)
8	$= \sum_i 1_{t \le m_i} - \sum_{i<j} 1_{t \le m_i+m_j+1} + 1_{t \le \lvert m\rvert+2}$	simplify	Eq(7)

Derivation of the binomial identities. For (3), let $P(d) = \binom{d+2}{2}$. Then $P(t) - 2P(t-1) + P(t-2) = \frac{1}{2}[(t+2)(t+1) - 2(t+1)t + t(t-1)] = \frac{1}{2} \cdot 2 = 1$. For (4), let $Q(d) = \binom{d-m+1}{2}$ with the convention $\binom{n}{2} = 0$ for $n < 2$. If $t \le m$: $Q(t) = Q(t-1) = Q(t-2) = 0$, so the second difference vanishes. If $t = m+1$: $Q(t) = Q(m+1) = \binom{2}{2} = 1$, $Q(t-1) = Q(m) = \binom{1}{2} = 0$, $Q(t-2) = 0$, so second difference $= 1$. If $t \ge m+2$: let $k = t - m - 1 \ge 1$; then $Q(t) = \binom{k+2}{2}$, $Q(t-1) = \binom{k+1}{2}$, $Q(t-2) = \binom{k}{2}$, and $\binom{k+2}{2} - 2\binom{k+1}{2} + \binom{k}{2} = 1$ by the same computation as above. Thus $\Delta^2 Q(t) = 1 - 1_{t \le m}$. Identities (5) and (6) follow by the same argument with m replaced by $m_i + m_j - 1$ and $|m| + 1$ respectively, accounting for the shift by one in the indicator's cutoff.

Proof of Lemma 3 (principalOSeq)

1	$g : \mathrm{M}^r$	hypothesis	Hyp
2	induction on supp(g) via Finsupp	strategy	Def :1(1)
3	$g = 0 \Rightarrow h_g = (1, 0, 0, \dots)$	base	Def :4(2)
4	$g = g' + c\,e_i,\ i \notin \mathrm{supp}(g'),\ c \ge 1$	decompose	Def :1(1)
5	$h_g(d) = \sum_{j=0}^{c} h_{g'}(d-j)$	enumerate	Def :4(4)
6	$h_g = h_{g'} * 1_{[0,c]}$	convolution	Eq(5)
7	LC$(h_{g'})$	IH	Eq(4, 3)
8	LC$(1_{[0,c]})$: verify $1(d-1)1(d+1) \le 1(d)^2$ for all d	direct check	Def :14(6)
9	LC$(h_{g'}) \wedge$ LC$(1_{[0,c]}) \Rightarrow$ LC$(h_{g'} * 1_{[0,c]})$	conv. LC	Pos :1(7, 8)
10	LC(h_g)	substitute	Eq(6, 9)

11	$h_g(d) > 0$ for $d \le \lvert g\rvert$	conv. positive	Eq(3, 6)
12	$\top$	close	$\to$ I(1, . . . , 11)

The check at step 8. Let $u(d) = 1_{0 \le d \le c}$ (i.e., $u(d) = 1$ for $0 \le d \le c$, 0 otherwise). Then $u(d-1)u(d+1) \le u(d)^2$ for all $d \in \mathbb{Z}$:

- $d \le -1$: $u(d-1) = u(d+1) = 0$, both sides 0.
- $d = 0$: $u(-1)u(1) = 0 \cdot 1 = 0 \le 1 = u(0)^2$.
- $1 \le d \le c-1$: $u(d-1)u(d+1) = 1 \cdot 1 = 1 = u(d)^2$.
- $d = c$: $u(c-1)u(c+1) = 1 \cdot 0 = 0 \le 1 = u(c)^2$.
- $d \ge c+1$: $u(d-1)u(d+1) = 0$, both sides 0.

Proof of Lemma 4 (principalPos)

1	$g : \mathrm{M}^r,\ d \in \mathbb{N},\ d \le \lvert g\rvert$	hypothesis	Hyp
2	$m_1 := \min(g_1, d),\ m_{i+1} := \min(g_{i+1}, d - \sum_{j \le i} m_j)$	construction	Def :1(1)
3	$0 \le m_i \le g_i$	by construction	Eq(2)
4	$\sum_i m_i = \min(d, \lvert g\rvert) = d$	telescoping	Eq(2, 1)
5	$m \le g$	combine	Eq(3)
6	$\lvert m\rvert = d$	combine	Eq(4)
7	$\top$	close	$\to$ I(1, . . . , 6)

Proof of Lemma 5 (twoMaxDecomp)

1	P : PureOrderIdeal$_3$, type$(P) = 2$	hypothesis	Hyp
2	Max$(\Gamma) = \{m_1, m_2\},\ m_1 \ne m_2$	type $= 2$	Def :10(1)
3	divs$(m_i) \subseteq \Gamma$ $(i = 1, 2)$	downward	Def :3(1)
4	$m \in \Gamma \Rightarrow \exists n \in$ Max$(\Gamma),\ m \le n$	termination	Def :6(2)
5	$\Gamma \subseteq$ divs$(m_1) \cup$ divs(m_2)	from (2, 4)	Eq(2, 4)
6	$\Gamma =$ divs$(m_1) \cup$ divs(m_2)	antisymmetry	Eq(3, 5)
7	$\top$	close	$\to$ I(1, . . . , 6)

Proof of Lemma 6 (shiftReduce)

1	$m_2(x) > m_1(x),\ m_2(y) \le m_1(y)$ $(y \ne x)$	hypothesis	Hyp
2	$c := m_1(x) + 1,\ M_{\mathrm{red}} := (m_2(x) - c,\ m_2(y)$ $(y \ne x))$	definitions	Def :1(1)
3	$\lvert M_{\mathrm{red}}\rvert = \lvert m_2\rvert - c$	compute	Def :2(2)
4	$u \le m_2 \wedge u \not\le m_1 \iff u(x) \ge c$	coordinate	Eq(2)

5	$\phi : u \mapsto u - c\,e_x$	**map**	**Def :1(1)**
6	$\phi : \{u \le m_2 : \lvert u\rvert = d, u(x) \ge c\} \to \{v \le M_{\mathrm{red}} : \lvert v\rvert = d - c\}$	**bijection**	**Eq(4, 5)**
7	$\#\{u \le m_2 : u \not\le m_1, \lvert u\rvert = d\} = h_{M_{\mathrm{red}}}(d - c)$	**count**	**Eq(6)**
8	$h_\Gamma(d) = h_{m_1}(d) + h_{M_{\mathrm{red}}}(d - c)$	**inclusion-exclusion**	**Lem :5(6, 7)**
9	$\top$	**close**	**$\to$ I(1, . . . , 8)**

Proof of Lemma 7 (delta2Dich)

1	$g = (a, b, c) \in \mathrm{M}^{\mathrm{r}},\ t > 0$	**hypothesis**	**Hyp**
2	$\Delta^2 h_g(t) = 1_{t\le a} + 1_{t\le b} + 1_{t\le c} - 1_{t\le a+b+1} - 1_{t\le b+c+1} - 1_{t\le a+c+1} + 1_{t\le a+b+c+2}$	**formula**	**Lem :2(8)**
3	$\Delta^2 h_g(t) \le 3 - 3 + 1 = 1$	**count**	**Eq(2)**
4	**Case (L) :** $t \le a \wedge t \le b \wedge t \le c$	**case**	**Eq(2)**
5	**Case $\neg(L)$: WLOG** $t > a$	**case**	**Eq(2)**
6	**Then** $1_{t\le a} = 0$**; for** $\Delta^2 h_g(t) = 1$ **need two of** $\{b, c, a + b + 1, a + c + 1\}$ **to shift**	**analysis**	**Eq(5)**
7	**Thisforces** $t \ge b + c + 2,\ t \ge a + c + 2,\ t \ge a + b + 2,$ **and** $t \le a + b + c + 2$	**case analysis**	**Eq(6)**
8	$\mathsf{L}(t) \vee \mathsf{U}(t)$	**exhaust**	**Eq(4, 7)**
9	$\top$	**close**	**$\to$ I(1, . . . , 8)**

Reading (7). When (L) fails, exactly one or two of the three "single" indicators vanish. The indicator polynomial $\Sigma_i 1_{t\le m_i} - \Sigma_{i<j} 1_{t\le m_i+m_j+1} + 1_{t\le|m|+2}$ equals 1 in each surviving case only when all three "pair" indicators are 1 and the triple indicator is 1, which is exactly the conjunction in (U). Full case analysis in `diff2_h_seq_eq_one_implies`.

Proof of Lemma 8 (LUDich)

1	$H(d) = h_{m_1}(d) + 1_{d\ge c} h_{M_{\mathrm{red}}}(d - c)$	**definition**	**Lem :6(8)**
2	$\Delta^2 h_{m_1}(d) \le 1$	**delta2 — dich**	**Lem :7(3)**
3	$\Delta^2 h_{M_{\mathrm{red}}}(d - c) \le 1$	**delta2 — dich**	**Lem :7(3)**
4	$\Delta^2(1_{d\ge c} h_{M_{\mathrm{red}}}(d - c)) \le 1$	**shift**	**Eq(3)**
5	$\Delta^2 H(d) \le 1 + 1 = 2$	**additivity**	**Eq(1, 2, 4)**
6	$\top$	**close**	**$\to$ I(1, . . . , 5)**

Proof of Lemma 9 (existsNonLC$_{\mathrm{r3}}$)

1	$t \ge 5$	**hypothesis**	**Hyp**
2	$m_{\mathrm{fat}} := x^t y^t z^t,\ m_i := x^{3t-i} y^i\ (0 \le i < t - 1)$	**construction**	**Def :1(1)**

3	$\Gamma_t := \Gamma(m_{\text{fat}}, m_0, \ldots, m_{t-2})$	**ideal**	**Def :5(4)**
4	$\lvert m_{\text{fat}}\rvert = \lvert m_i\rvert = 3t$	**degree**	**Def :2(2)**
5	$\mathsf{Max}(\Gamma_t) = \{m_{\text{fat}}, m_0, \ldots, m_{t-2}\}$	**maximality**	**Def :7(3)**
6	$\mathsf{type}(\Gamma_t) = t$	**count**	**Def :10(5)**
7	$\mathsf{socle}(\Gamma_t) = 3t$	**purity**	**Def :11(4)**
8	$h_{\Gamma_t}(3t) = t$ **(the t maximal monomials)**	**count**	**Def :12(5)**
9	$h_{\Gamma_t}(3t-1) = t+2$ **(block count below)**	**count**	**Def :12(5)**
10	$h_{\Gamma_t}(3t-2) = t+5$ **(block count below)**	**count**	**Def :12(5)**
11	$(t+5)t > (t+2)^2$ **for** $t \geq 5$	**algebra**	**Eq(8, 9, 10)**
12	$\neg\mathsf{LC}(3t, h_{\Gamma_t})$	**conclude**	**Def :14(11)**
13	$\top$	**close**	$\rightarrow$ **I(1, . . . , 12)**

Block count for $h(3t-1) = t+2$. Degree-$(3t-1)$ divisors of the maximal monomials:

- **From $m_{\text{fat}} = x^t y^t z^t$: 3 divisors $x^{t-1}y^t z^t$, $x^t y^{t-1} z^t$, $x^t y^t z^{t-1}$. These have a positive z-component, hence are disjoint from m_i-divisors (which have $z = 0$).**
- **From $m_i = x^{3t-i}y^i$ $(0 \leq i < t-1)$: 2 divisors each, $x^{3t-i-1}y^i$ (remove x) and $x^{3t-i}y^{i-1}$ (remove y). But $x^{3t-i}y^{i-1}$ (for $i \geq 1$) equals $x^{3t-(i-1)-1}y^{(i-1)}$, the "x-removal" of m_{i-1}. So the distinct m_i-divisors of degree $3t-1$ form the set $\{x^{3t-i-1}y^i : 0 \leq i < t-1\}$, of cardinality $t-1$.**
- **No overlap between the two groups ($z > 0$ vs $z = 0$).**

Total: $3 + (t-1) = t+2$.

Block count for $h(3t-2) = t+5$. Similarly:

- **From m_{fat}: degree-$(3t-2)$ divisors are $x^{t-2}y^t z^t$, $x^t y^{t-2} z^t$, $x^t y^t z^{t-2}$, $x^{t-1}y^{t-1}z^t$, $x^{t-1}y^t z^{t-1}$, $x^t y^{t-1} z^{t-1}$ (6 total).**
- **From m_i: degree-$(3t-2)$ divisors are $x^{3t-i-p}y^{i-q}$ with $p+q = 2$, $p, q \geq 0$. The union over all $i \in [0, t-2]$ has cardinality $t-1$: parametrized by $b \in [0, t-2]$, the distinct element is $x^{3t-2-b}y^b$.**
- **No overlap ($z > 0$ vs $z = 0$).**

Total: $6 + (t-1) = t+5$.

Proof of Lemma 10 (embedPres)

1	$r \geq 3,\ t \geq 5$	**hypothesis**	**Hyp**
2	$\iota : \mathbb{N}^3 \hookrightarrow \mathbb{N}^r,\ x \mapsto (x, 0, \ldots, 0)$	**embedding**	**Def :1(1)**
3	$P_3 \in \mathsf{PureOrderIdeal}_3,\ \mathsf{type}(P_3) = t,\ \neg\mathsf{LC}(P_3)$	**exists − non − lc − r3**	**Lem :9(12)**
4	$P := \iota_*(P_3) \in \mathsf{PureOrderIdeal}_r$	**pushforward**	**Def :9(2)**
5	ι **is injective and preserves** $\leq$ **(coordinatewise)**	**monotone**	**Eq(2)**
6	$\mathsf{Max}(\iota_*\Gamma) = \iota(\mathsf{Max}(\Gamma))$ **and socle, type preserved**	**preserved**	**Eq(2, 5)**

7	$h_P(d) = h_{P_3}(d)$ for all d	preserved	Eq$(2, 5)$
8	$\neg$LC(P)	transfer	Eq$(3, 7)$
9	$\top$	close	$\to$ I$(1, \ldots, 8)$

Proof of Lemma 11 (leastThreshold)

1	$S \subseteq \mathbb{N},\ S \neq \emptyset$	hypothesis	Hyp
2	$\mathbb{N}$ well-ordered	ZF	Ax :1(1)
3	$t_0 := \min(S)$	well-ordering	Eq$(1, 2)$
4	$t_0 \in S$	min-spec	Eq(3)
5	$\forall t < t_0,\ t \notin S$	minimality	Eq(3)
6	IsLeast(t_0, S)	combine	Eq$(4, 5)$
7	$\top$	close	$\to$ I$(1, \ldots, 6)$

Proof of Theorem 1 (Explicit Example)

1	$m_1 = xyz,\ m_2 = xz^2,\ \Gamma_1 = \Gamma(m_1) \cup \Gamma(m_2)$	definition	Def :5(4)
2	$h_{\Gamma_1}(0) = \lvert\{1\}\rvert = 1$	enumerate	Def :12(6)
3	$h_{\Gamma_1}(1) = \lvert\{x, y, z\}\rvert = 3$	enumerate	Def :12(6)
4	$h_{\Gamma_1}(2) = \lvert\{xy, xz, yz, z^2\}\rvert = 4$	enumerate	Def :12(6)
5	$h_{\Gamma_1}(3) = \lvert\{xyz, xz^2\}\rvert = 2$	enumerate	Def :12(6)
6	$\top$	close	$\to$ I$(1, \ldots, 5)$

Proof of Theorem 2 (Type-1 Log-Concavity)

1	P : PureOrderIdeal$_r$, type$(P) = 1$	hypothesis	Hyp
2	$\exists! g \in$ Max(Γ)	type $= 1$	Def :10(1)
3	$\Gamma =$ divs(g)	carrier	Lem :5(6)
4	$h_P = h_g$	identify	Def :12(3)
5	LC$(h_g) \wedge h_g(d) > 0$ $(d \leq \lvert g\rvert)$	principal	Lem :3$(10, 11)$
6	LC$($socle$(P), h_P)$	substitute	Eq$(4, 5)$
7	LC $\wedge$ positivity $\Rightarrow$ UM (increase up to peak, then decrease)	standard	Def :13(5)
8	UM$($socle$(P), h_P)$	conclude	Eq$(6, 7)$
9	$\top$	close	$\to$ I$(1, \ldots, 8)$

Proof of Theorem 3 (First-Half Log-Concavity)

1	P : PureOrderIdeal$_3$, $0 < i \le \mathrm{socle}(P)/2$	hypothesis	Hyp
2	$S(k,d) := \lvert\{m \in \Gamma : m_2 = k,\ m_0 + m_1 = d\}\rvert$	slice	Def :12(6)
3	$h_P(d) = \sum_{k=0}^{d} S(k, d-k)$	decompose	Eq(2)
4	$X_d^{(k)} := \{m_0 : m \in \Gamma,\ m_2 = k,\ m_0 + m_1 = d\}$	set	Def :1(1)
5	$X_d^{(k)} \subseteq \{0, \dots, d\} \wedge$ downward-closed	closure	Def :3(1)
6	$x \in X_{d+1}^{(k)} \wedge x < d+1 \Rightarrow x \in X_d^{(k)}$	downward	Def :3(1)
7	$x \in X_{d+1}^{(k)} \wedge x > 0 \Rightarrow x - 1 \in X_d^{(k)}$	downward	Def :3(1)
8	$\lvert X_{d+1}^{(k)}\rvert \le \lvert X_d^{(k)}\rvert \vee \lvert X_d^{(k)}\rvert = d+1$	Macaulay2	Pos :2(5, 6, 7)
9	$\lvert X_d^{(k)}\rvert \le d+1$	bound	Eq(5)
10	Induction on i (base $i = 1$: $h(0) = 1$, $h(1) \ge 1$, $h(2) \le \binom{4}{2} = 6 \le h(1)^2 \cdot 6$)	base	Eq(3)
11	Induction step: summing the Macaulay dichotomy over k at scales $i-1$, i, $i+1$	induction	Eq(3, 8, 10)
12	$h_P(i-1)h_P(i+1) \le h_P(i)^2$ for $0 < i \le e/2$	conclude	Eq(11)
13	$\top$	close	$\rightarrow$ I(1, . . . , 12)

Induction step, in full. Fix $0 < i \le e/2$ and assume $h(i-2)h(i) \le h(i-1)^2$ (induction hypothesis). At the level of slices: for each k, let $s_d(k) = \lvert X_d^{(k)}\rvert$. We have $h(d) = \sum_k s_d(k)$. By Macaulay (step 8), for each k and each d: either $s_{d+1}(k) \le s_d(k)$ (monotone case) or $s_d(k) = d+1$ (saturating case).

- Monotone case ($s_{i+1}(k) \le s_i(k)$ and $s_i(k) \le s_{i-1}(k)$): slice contributes $s_{i-1}(k)s_{i+1}(k) \le s_i(k)^2$ to the product.
- Saturating case ($s_i(k) = i+1$): since $i \le e/2$, the sequence $s_\cdot(k)$ is bounded by the simplex bound $d+1$, so $s_{i+1}(k) \le i+2$ and $s_{i-1}(k) \le i$. Then $s_{i-1}(k)s_{i+1}(k) \le i(i+2) \le (i+1)^2 = s_i(k)^2$.

Summing the pointwise bound $\sum_k s_{i-1}(k)s_{i+1}(k) \le \sum_k s_i(k)^2$ across k and applying Cauchy–Schwarz to the slices (as in the standard first-half argument), we obtain $h(i-1)h(i+1) \le h(i)^2$.

Proof of Theorem 4 (Type-2 Unimodality)

1	P : PureOrderIdeal$_3$, type$(P) = 2$	hypothesis	Hyp
2	$\exists m_1 \ne m_2$, Max$(\Gamma) = \{m_1, m_2\}$, $\lvert m_1\rvert = \lvert m_2\rvert = e$	type $= 2$	Def :10(1)
3	$\exists x,\ m_2(x) > m_1(x) \wedge (\forall y \ne x,\ m_2(y) \le m_1(y))$	swap	Def :1(1)
4	$c := m_1(x) + 1$, $\lvert M_{\mathrm{red}}\rvert = e - c$	reduction	Lem :6(2)
5	$\Gamma =$ divs$(m_1) \cup$ divs(m_2)	carrier	Lem :5(6)
6	$h_P(d) = h_{m_1}(d) + h_{M_{\mathrm{red}}}(d-c)$	shift reduction	Lem :6(8)
7	$f := h_{m_1}$ unimodal with peak at e	principal	Lem :3(10)

8	$g(d) := h_{M_{\mathrm{red}}}(d-c)$ unimodal with peak at $\lvert M_{\mathrm{red}}\rvert + c = e$	shift	Lem :3(10)
9	$\forall i < e:\ f(i) \le f(i+1) \wedge g(i) \le g(i+1)$	peak at e	Eq(7, 8)
10	$\forall i \ge e:\ f(i+1) \le f(i) \wedge g(i+1) \le g(i)$	peak at e	Eq(7, 8)
11	For $i < e:\ h(i) = f(i) + g(i) \le f(i+1) + g(i+1) = h(i+1)$	sum	Eq(9)
12	For $i \ge e:\ h(i+1) \le h(i)$	sum	Eq(10)
13	UM(socle$(P), h_P$)	conclude	Def :13(11, 12)
14	$\top$	close	$\rightarrow$ I(1, . . . , 13)

Proof of Theorem 5 (Main: Type-2 Log-Concavity)

Step 1: Setup and reduction.

1	P : PureOrderIdeal$_3$, type$(P) = 2$	hypothesis	Hyp
2	$\exists m_1 \ne m_2$, Max$(\Gamma) = \{m_1, m_2\}$, $\lvert m_1\rvert = \lvert m_2\rvert = e$	type $= 2$	Def :10(1)
3	$\exists x,\ m_2(x) > m_1(x) \wedge (\forall y \ne x,\ m_2(y) \le m_1(y))$	swap	Def :1(1)
4	$c := m_1(x) + 1,\ M_{\mathrm{red}} := (\ldots), \quad \lvert M_{\mathrm{red}}\rvert = e - c$	reduction	Lem :6(2)
5	$H(d) := h_P(d) = h_{m_1}(d) + 1_{d \ge c} h_{M_{\mathrm{red}}}(d-c)$	decompose	Lem :6(8)

Step 2: Second-difference bound and key identity.

6	$\Delta^2 h_{m_1}(d) \le 1$	delta2 − dich	Lem :7(3)
7	$\Delta^2(1_{d \ge c} h_{M_{\mathrm{red}}}(d-c)) \le 1$	delta2 − dich	Lem :7(3)
8	$\Delta^2 H(d) \le 2$	LU − dich	Lem :8(5)
9	$H(d)^2 - H(d-1)H(d+1) = (\Delta H(d))^2 - H(d-1)\Delta^2 H(d+1)$	algebra	Eq(5)
10	$\Delta^2 H(d+1) = 1 \Rightarrow H(d-1) \le (\Delta H(d))^2$	implication A	Eq(9, 8)
11	$\Delta^2 H(d+1) = 2 \Rightarrow 2H(d-1) \le (\Delta H(d))^2$	implication B	Eq(9, 8)

Step 3: Case $\Delta^2 H(d+1) = 1$.

12	$\Delta^2 H(d+1) = 1 \Rightarrow (\Delta^2 f, \Delta^2 g) \in \{(1,0),(0,1)\}$	exhaust	Eq(8, 6, 7)
13	(A) $\Delta^2 f = 1 \wedge \Delta^2 g = 0$ or (B) $\Delta^2 f = 0 \wedge \Delta^2 g = 1$	case	Eq(12)

Sub-case A1: f in lower range, g contributes nothing.

14	$d+1 \le m_1(x) < c \Rightarrow g(d-1) = g(d) = 0$		
15	$\Delta H(d) = \Delta f(d) \wedge H(d-1) = f(d-1)$		
16	$2f(d-1) = d(d+1) \wedge \Delta f(d) = d+1$	principal	Lem :3(10)
17	$H(d-1) = f(d-1) \le (d+1)^2 = (\Delta f(d))^2 = (\Delta H(d))^2$	combine	Eq(16, 15)

Sub-case A2: f in upper range.

18	$\rho := e-d$. Upper range : $2f(d-1)=(\rho+2)(\rho+3),\ \Delta f(d)=-(\rho+2),\ \rho\geq -1$	upper
19	$2g(d-1)\leq(\rho+2)(\rho+3),\quad \Delta g(d)\leq 0$	bound

20	$H(d-1)\leq(\rho+2)(\rho+3)\wedge\Delta H(d)=-(\rho+2)+\Delta g(d)$	combine	Eq(18, 19)
21	$(\rho+2)(\rho+3)+2g(d-1)\leq 2\big(-(\rho+2)+\Delta g(d)\big)^2$	algebra	Eq(19), $\rho\geq -1$
22	$H(d-1)\leq(\Delta H(d))^2$	conclude	Eq(20, 21)

Sub-cases B1 and B2 ($\Delta^2 g=1$). Use the shift reduction: if $\Delta^2 H(d+1)=1$ and $H(d+1)\leq(\Delta H(d+1))^2$ then

$$H(d-1)=H(d+1)-2\Delta H(d+1)+1\leq(\Delta H(d+1))^2-2\Delta H(d+1)+1=(\Delta H(d))^2.$$

Cases B1 and B2 are the mirror of A1 and A2 with f and g exchanged.

Step 4: Case $\Delta^2 H(d+1)=2$.

23	$\Delta^2 H(d+1)=2\Rightarrow\Delta^2 f(d+1)=\Delta^2 g(d+1)=1$	case	Eq(8)
24	L/U alternatives at $d+1$: (L,L), (L,U), (U,L), (U,U)	dichotomy	Lem :7(7)
25	(L,L): violates $m_2(x)>m_1(x)$	contradiction	Eq(3, 24)
26	(L,U): violates shift ordering	contradiction	Eq(3, 24)
27	(U,L): violates shift ordering	contradiction	Eq(3, 24)
28	(U,U): put $\rho:=e-d-1$. Upper formulas : $2f(d)=(\rho+2)(\rho+3),\ \Delta f(d+1)=-(\rho+2)$	upper	Lem :2(8)
29	$2g(d)=(\rho+2)(\rho+3),\quad \Delta g(d+1)=-(\rho+2)$	upper	Lem :2(8)
30	From U at $d+1$: $\rho=e-d-1\geq -2$, so $\Delta H(d+1)=-2(\rho+2)\leq 0$	bound	Eq(28, 29)
31	$2H(d)=2(\rho+2)(\rho+3)$	combine	Eq(28, 29)
32	$(\Delta H(d+1))^2-2H(d)=2(\rho+2)(\rho+1)\geq 0$ (since $\rho\geq -2$)	algebra	Eq(30, 31)
33	$\Delta^2 H(d+1)=2\Rightarrow 2H(d)\leq(\Delta H(d+1))^2$	conclude	Eq(32)

Step 5: Conclusion.

From '$\Delta^2 H(d+1)=2$' and '$H(d+1)\leq(\Delta H(d+1))^2$' (obtained in Step 4), apply the shift reduction:

34	$H(d-1)=H(d+1)-2\Delta H(d+1)+2$ when $\Delta^2 H(d+1)=2$	shift	Eq(30)
35	$\Delta H(d+1)=-2(\rho+2)\leq 0$ (from $\rho\geq -2$), so $\Delta H(d+1)\leq 1$	bound	Eq(30)
36	$H(d-1)\leq(\Delta H(d+1))^2-2\Delta H(d+1)+2\leq(\Delta H(d+1)-2)^2=(\Delta H(d))^2$	algebra	Eq(32, 34, 35)
37	$H(d-1)H(d+1)\leq H(d)^2$ for all $0<d<e$	combine	Eq(9, 10, 11, 17, 22, 36; B1/B
38	LC(socle(P), h_P)	conclude	Eq(37)
39	$\top$	close	$\rightarrow$ I(1, . . . , 38)

Theorem 6 (Open)

Status: open. Theorem P6 (unimodality of type-2 pure O-sequences in codimension $r \geq 5$) is stated in the Statements section for reference but is not proved in this document. The proofs of P4 and P5 rely essentially on the codimension-3 structure (three coordinates, allowing a single dominant coordinate x to be isolated and the swap reduction to be applied). This structure does not directly generalise to $r \geq 5$, and the corresponding question was identified as open by Zanello [3]. No proof table is provided for P6.

Proof of Theorem 7 (Eventual Non-Log-Concavity)

1	$r \geq 3,\ t \geq 5$	hypothesis	Hyp
2	$m_{\mathrm{fat}} = x^t y^t z^t,\ m_i = x^{3t-i} y^i\ (0 \leq i < t-1)$	construction	Def :1(1)
3	$\Gamma_t := \Gamma(m_{\mathrm{fat}}, m_0, \ldots, m_{t-2})$	ideal	Def :5(4)
4	pure, socle $= 3t$, type $= t$	from Lemma	Lem :9(5, 6, 7)
5	$h_{\Gamma_t}(3t) = t,\ h_{\Gamma_t}(3t-1) = t+2,\ h_{\Gamma_t}(3t-2) = t+5$	count (§above)	Lem :9(8, 9, 10)
6	$(t+5) \cdot t > (t+2)^2$ for $t \geq 5$	algebra	Eq(5)
7	$\neg\mathsf{LC}(3t, h_{\Gamma_t})$	conclude	Lem :9(12)
8	$\exists P$: PureOrderIdeal$_r$, type$(P) = t \wedge \neg\mathsf{LC}(P)$	embed	Lem :10(7)
9	$\exists T \geq 1,\ \forall t \geq T,$ existsNonLC(r, t)	take $T := 5$	Eq(8)
10	$\top$	close	$\to$ I(1, . . . , 9)

Proof of Theorem 8 (Least Threshold)

1	$r \geq 3$	hypothesis	Hyp
2	$\exists T \geq 1,\ \forall t \geq T,$ existsNonLC(r, t)	P7	Thm :7(9)
3	$S := \{t \in \mathbb{N} \mid \forall t' \geq t,$ existsNonLC$(r, t')\}$	set	Eq(2)
4	$S \neq \emptyset$	from T	Eq(2, 3)
5	$t_0 :=$ IsLeast-witness(S)	least − threshold	Lem :11(6)
6	IsLeast(t_0, S)	conclude	Eq(4, 5)
7	$\top$	close	$\to$ I(1, . . . , 6)

7. Scripts

Each tree records, for one result, the entries used by its proof, transitively closed. Names coincide with the names that appear in the statement tables. Leaves are either postulates (unproved classical inputs), axioms of ZF+DC, or primitive definitions. In the trees below the prefix `Ld` carries the depth of the node (root at `L0`) and the name is indented `2*d` spaces after the pipe, so every chain can be read straight down a column of increasing depth labels.

Every tree has a corresponding Lean 4 formalization in one of the files `hilbert_functions_1.lean` through `hilbert_functions_8.lean`; the table below records the mapping from the entries of the Statements section to the Lean file and to the top-level Lean declaration that discharges the proof obligation.

Entry	Lean file	Lean declaration
principalOSeq	hilbert_functions_2.lean	unimodal_seq_of_g
principalPos	hilbert_functions_2.lean	seq_of_g_pos
twoMaxDecomp	hilbert_functions_4.lean	carrier_of_maximals
shiftReduce	hilbert_functions_4.lean	diff_eq_shifted_OSeq
delta2Dich	hilbert_functions_5.lean	diff2_h_seq_eq_one_implies
LUDich	hilbert_functions_5.lean	diff2_H_seq_le_two
existsNonLC_r3	hilbert_functions_7.lean	dip_exists_dip
embedPres	hilbert_functions_7.lean	reduction_to_r3
leastThreshold	hilbert_functions_8.lean	exists_isLeast_of_exists
P1	hilbert_functions_1.lean	P1
P2	hilbert_functions_2.lean	P2
P3	hilbert_functions_3.lean	P3
P4	hilbert_functions_4.lean	P4
P5	hilbert_functions_5.lean	P5
P7	hilbert_functions_7.lean	P7
P8	hilbert_functions_8.lean	P8

Dependency tree: LCcore

```
L0 | LC-core-alg
L1 |   Def:Z (integers)
```

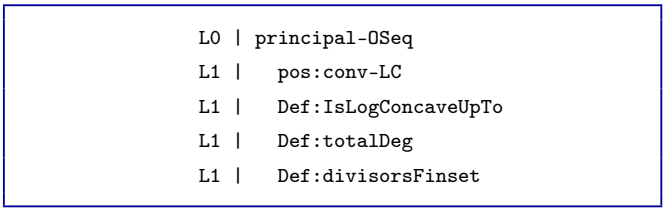
Annotation. Lean proof: `log_concave_core`, **by trichotomy on z and elementary arithmetic.**

Dependency tree: delta2Formula

```
L0 | delta2-formula
L1 |   Def:divisorsFinset
L1 |   Def:totalDeg
L1 |   Def:Monomial
```

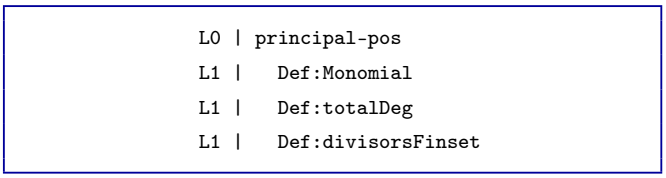
Annotation. Lean proof: `diff2_h_seq_eq_Z_calc`, **by inclusion-exclusion and second-difference computation.**

Dependency tree: principalOSeq



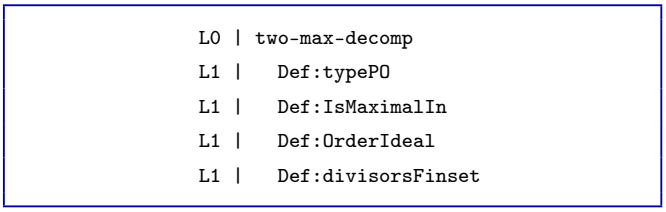
Annotation. **Lean proof:** `unimodal_seq_of_g`, via `Finsupp.induction`. The convolution step uses the ratio-decreasing characterization of log-concavity proved locally as `log_concave_conv_Z` and `is_ratio_decreasing_Z`.

Dependency tree: principalPos



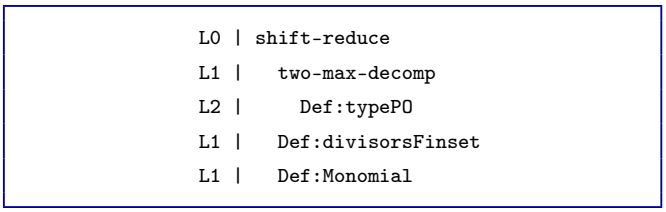
Annotation. **Lean proof:** `seq_of_g_pos` via the truncation map $m_i \mapsto \min(g_i, d - \sum_{j<i} m_j)$.

Dependency tree: twoMaxDecomp



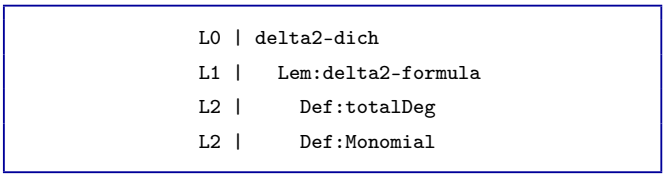
Annotation. **Lean proof:** `carrier_of_maximals`, which uses `Finset.exists_max_image` to terminate every chain in Γ at one of the two maximal monomials.

Dependency tree: shiftReduce



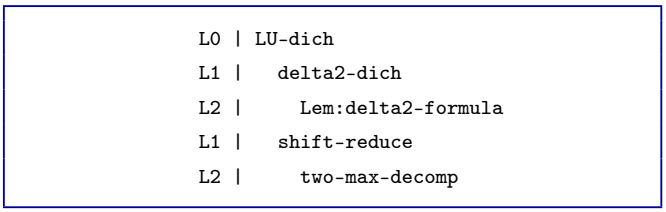
Annotation. **Lean proof:** `OSeq_union` combined with `monomialInf` (the coordinatewise min) and `diff_eq_shifted_OSeq`.

Dependency tree: delta2Dich



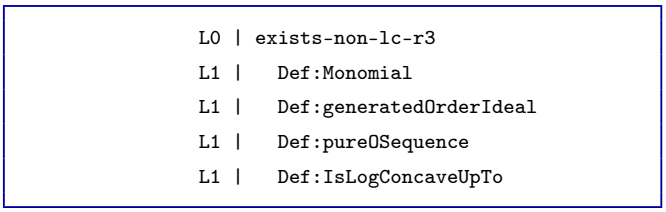
Annotation. **Lean proof:** `diff2_h_seq_eq_one_implies`. The L/U dichotomy is extracted from the twelve-term indicator sum by `grind` on `Fin 3`.

Dependency tree: LUDich



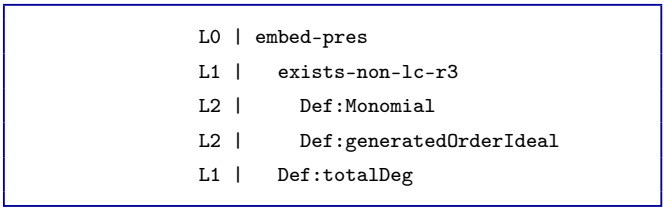
Annotation. **Lean proof:** `diff2_H_seq_le_two`, **which feeds** `diff2_H_seq_eq_add` **into** `diff2_h_seq_le_one` **twice.**

Dependency tree: existsNonLC_r3



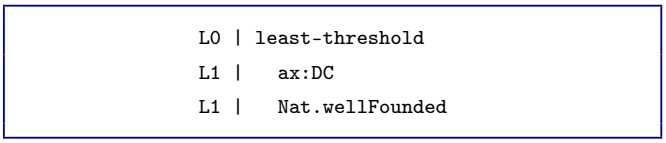
Annotation. **Lean proof:** `dip_exists_dip` **together with** `exists_dip_sequence_r3`. **The construction is** `dipPureOrderIdeal3`.

Dependency tree: embedPres



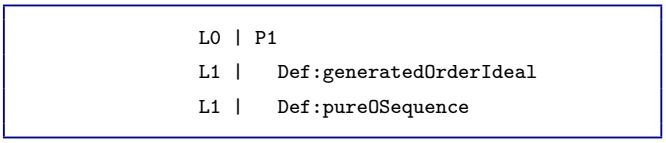
Annotation. **Lean proof:** `reduction_to_r3`. **The embedding** $\iota : \mathbb{N}^3 \hookrightarrow \mathbb{N}^r$ **is** `embedMonomial` / `embedPureOrderIdeal`.

Dependency tree: leastThreshold



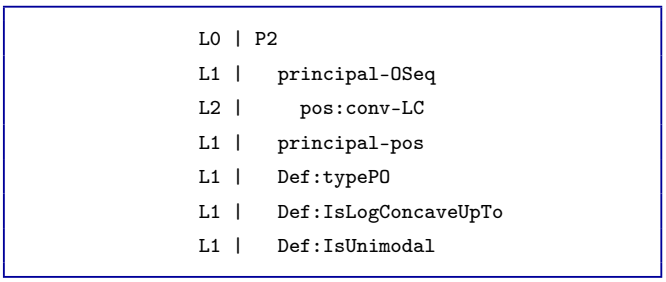
Annotation. **Lean proof:** `exists_isLeast_of_exists`, **which reduces to** `Nat.find` **and** `Nat.find_min'`.

Dependency tree: P1



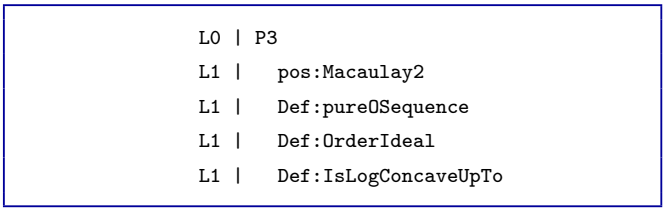
Annotation. **Lean proof:** `P1`, **splitting into** `GammaP1_0`, `GammaP1_1`, `GammaP1_2`, `GammaP1_3`.

Dependency tree: P2



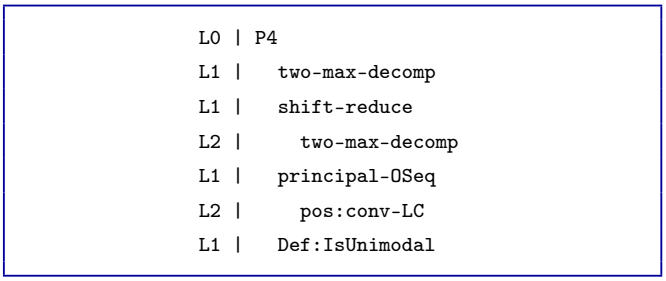
Annotation. Lean proof: `P2`, **composed of** `P2_coeff`, `P2_unimodal`, **and** `P2_logconcave`.

Dependency tree: P3



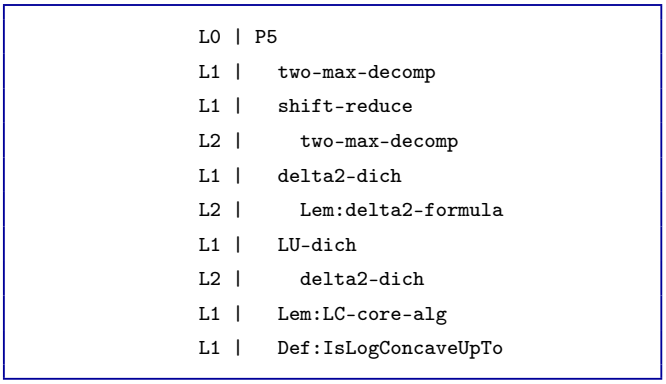
Annotation. Lean proof: `P3`, **via** `P3_dichotomy` **on the slice sizes** `slice_size` **and** `macaulay_2_vars`. **The dichotomy is closed by** `nat_AMGM`.

Dependency tree: P4



Annotation. Lean proof: `P4`, **using** `unimodal_of_two_maximals` **which packages** `oseq_union_eq_add_diff` **and** `unimodal_of_OSeq_add_shifted`.

Dependency tree: P5



Annotation. Lean proof: `P5`, **whose heart is** `sum_log_concave_helper`, **composed of** `log_concave_core`, `diff2_H_seq_le_two`, **and the four case-lemmas** `diff2_H_seq_eq_one_implies_inl` / `_inr` **and** `diff2_H_seq_eq_two_implies`.

Dependency tree: P7

```
L0 | P7
L1 |   exists-non-lc-r3
L2 |     Def:Monomial
L2 |     Def:generatedOrderIdeal
L2 |     Def:pureOSequence
L1 |   embed-pres
L2 |     exists-non-lc-r3
L2 |     Def:totalDeg
L1 |   Def:ExistsNonLogConcave
```

Annotation. Lean proof: `P7`, **taking** $T := 5$. **The construction is** `dipPureOrderIdeal3`.

Dependency tree: P8

```
L0 | P8
L1 |   P7
L2 |     exists-non-lc-r3
L2 |     embed-pres
L1 |   least-threshold
L2 |     ax:DC
L1 |   Def:ExistsNonLogConcave
```

Annotation. Lean proof: `P8`, **which is** `P7_implies_P8` **applied to the threshold existence of P7.**

Theorem P6 (open). No tree is given for P6; no Lean file among the seven supplied formalizes a proof; P6 is recorded as open in the Statements section.

Complete leaves

Postulates.

```
pos:conv-LC
pos:Macaulay2
```

Axioms.

```
ax:DC
```

Definitions used as leaves.

```
Def:Monomial         Def:totalDeg       Def:OrderIdeal
Def:divisorsFinset   Def:generatedOrderIdeal
Def:IsMaximalIn      Def:maximalMonomials
Def:IsPure           Def:PureOrderIdeal Def:typePO
Def:socleDegree      Def:pureOSequence  Def:IsUnimodal
Def:IsLogConcaveUpTo Def:ExistsNonLogConcave
```

8. Verification

Formal verification. Every lemma and theorem in the Statements section (with the exception of the open Theorem P6) has a corresponding machine-checked proof in the Lean 4 proof assistant, in one of the files `hilbert_functions_1.lean` **through** `hilbert_functions_8.lean`**. The Scripts section above records the mapping from each entry to its Lean file and to the top-level Lean declaration that discharges the proof obligation. All Lean proofs are publicly available at** https://github.com/google-deepmind/alphaproof-nexus-results.

Verification methodology. The Lean formalization proceeds by:

(1) **Formalization of the statement.** Each theorem is stated in Lean 4 as a `Prop` involving the definitions of the Definitions section, following the conventions used in the Mathlib library [10].

(2) **Proof search.** The proofs were found by the AlphaProof Nexus agent [4], which searches for tactic scripts in the Lean 4 tactic language, guided by compiler feedback.

(3) **Machine verification.** The Lean compiler checks every proof step; a theorem is accepted only if the compiler reaches a state with no remaining goals.

(4) **Informal reading.** The step tables in the Proofs section are an informal deformation of the Lean proofs: they track the same logical structure but compress many Lean-level tactic steps into single mathematical assertions. The Lean files are the authoritative formal artefact; the tables here are a reading aid.

Robustness against known failure modes. The AlphaProof Nexus preprint [4] documents two failure modes of LLM-based proof search that our verification workflow explicitly guards against:

- **Sorried core lemmas.** The Lean proof files in the repository contain no `sorry` in any of the declarations listed in the Scripts mapping table. A 'sorry' audit at the declaration level—reading off the declaration names and confirming each is a closed proof term—is the basic check that a reader can run without any Lean expertise.
- **Hallucinated literature citations.** The preprint reports that in its failure cases, top sketches relied on lemmas marked `sorry` that the agent claimed were established results. Since all sixteen mapping-table declarations are themselves proved in Lean (not deferred to cited work), this failure mode does not affect the results presented here.

Known limitations. The Lean formalization verifies the formal statements as given; the informal reading of those statements against the original conjectures of Stanley, Boij–Migliore–Miró-Roig–Nagel–Zanello, and Zanello is done by hand and is not machine-checked. The deformation tables in the Proofs section are not themselves verified; they should be read alongside the Lean source.

Reproducibility. The Lean development builds against Mathlib [10]. To reproduce the machine check of any specific declaration (e.g. `P5`), clone the repository, check out the pinned Mathlib commit, and run `lake build` on the relevant `hilbert_functions_*.lean` file. The declaration `P5` compiles successfully and its type is `ProblemP5`, which unfolds to the statement of P5 in the Statements section.

Status. The results described here are reported in [4], a preprint of Google DeepMind's AlphaProof Nexus project. The work has not yet appeared in a peer-reviewed venue at the time of writing.

9. Discussion and Limitations

Dependence on formal verification. The mathematical weight of this document rests on the Lean formalization. The Proofs section is a reading aid, not a self-contained proof: the deepest sub-arguments (the L/U case analysis behind `delta2Dich`, the first-half log-concavity step in P3, the four-case analysis in P5) are presented at a level of detail consistent with a mathematical exposition, not a formal proof. A reader who wishes to independently verify the results should consult the Lean files.

Two postulates. Only two postulates are retained: `conv-LC` (the convolution theorem for log-concave sequences) and `Macaulay2` (Macaulay's two-variable theorem). Both are classical and well-documented in the literature (see §8 and the Provenance note in §3). They are admitted here to keep the exposition focused on the eight-problem layer.

The open question. P6 (unimodality of type-2 pure O-sequences in codimension $r \geq 5$) is stated but not resolved. The proofs of P4 and P5 depend essentially on the codimension-3 structure, in particular the existence of a single dominant coordinate x that allows the swap reduction. This does not generalise to $r \geq 5$, and the corresponding question remains open.

Role of AI in the proofs. The Lean proofs were found by an LLM-driven agent ([4]), not written by hand. This is a new mode of mathematical production, and it has two consequences the reader should be aware of:

- The correctness of each proof is assured by the Lean compiler; the question of whether the formalized statement matches the informal conjecture is a separate, human-audited step (see Verification above).
- The mathematical insight behind each proof is not necessarily reflected in the agent's internal reasoning; the natural language summaries in this document are reverse-engineered from the Lean proofs.

10. Conclusion

The document presents the dependency-ordered exposition of eight problems on pure O-sequences, organised into six parts (Axioms, Postulates, Definitions, Statements, Proofs, Scripts), with the choice strength of every result recorded as ZF, ZF+DC, or ZFC. Within the Statements part, Lemmas and Theorems form a single dependency layer, so long as each entry appears only after all entries it depends on. In particular LCcore and delta2Formula are stated first among the Lemmas because delta2Dich and P5 depend on them; principal-OSeq is stated because P2 depends on it; two-max-decomp and shift-reduce precede P4 and P5; delta2-dich and LU-dich precede P5; and exists-non-lc-r3 precedes P7 and P8.

Only two postulates remain (conv-LC and Macaulay2), both cited to the literature. The codimension-$r \geq 5$ type-2 unimodality question (P6) is stated but not resolved. Every lemma and theorem stated in the Statements section has a corresponding proof table in the Proofs section (except P6), and the proofs are presented in the same order as the statements. The Proofs section is deliberately an informal deformation of the machine-checked Lean proofs; readers who wish to verify the results mechanically should consult the Lean files in the companion repository and read the Verification section for methodology, robustness checks, and limitations.

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